Confidential Instructions:

Candidates are advised to spend no more than:

- **60 minutes** on **Question 1**.
- **60 minutes** on **Question 2**.
- **30 minutes** on **Question 3**.

Preparation of materials

Question 1

i. 1.0 mol dm–3 sucrose solution may be prepared the day before the examination.
It should be kept in a covered container in a refrigerator.
0.5% methylene blue solution may be prepared the day before the examination.
It should be kept in a covered container.
The solutions must be at **room temperature** for the examination.

ii. **S**, 1.0 mol dm–3 sucrose solution
This is prepared by sprinkling 68.4 g of sucrose, a little at a time, onto the surface of 80 cm3 of distilled water, stirring continuously as you sprinkle. Make up to 200 cm3 with distilled water.

iii. **P**, at least 7 pieces of peeled potato wrapped in a damp paper towel in a covered dish, labelled **P.**
You may use any variety of the white (or Irish) potato, *Solanum tuberosum*.
Cut each piece of potato with a cross-sectional area of 1.5 cm × 1 cm.
Each candidate should be provided with a mixture of different lengths, varying from 4.5 cm to 6 cm.
The potato pieces for each candidate should be prepared on the day of the examination.

iv. **M**, 0.5% methylene blue solution
This is prepared by putting 0.5 g of methylene blue into 80 cm3 of distilled water and stirring continuously. Make up to 100 cm3 with distilled water.

Question 2

i. 15 cm^3 of 0.1% 2,6-dichlorophenol indophenol (DCPIP) solution in a corked, labelled container (labelled as **DCPIP**). Dissolve powder in distilled water.

ii. 5 cm^3 of a standard solution of ascorbic acid (labelled as **vitamin C solution**) prepared as follows:
Dissolve 400 mg of ascorbic acid powder in 100 cm^3 distilled water. Add 15 drops of BDH Universal Indicator solution and adjust the pH to between 7 and 8 by adding 5% sodium hydroxide solution drop by drop. This is **4.0mgcm^{-3}** ascorbic acid solution.
The solution should be dispensed to students in a corked, **labelled** tube from which 1cm^3 volumes can be withdrawn.

iii. 5 cm3 each of fresh lemon juice, orange juice, grapefruit juice. Dispense to students in corked tubes **labelled** lemon juice, orange juice and grapefruit juice.

Apparatus List

Each candidate will require:

materials and apparatus for each candidate	quantity
1.0 mol dm^{-3} sucrose solution in a beaker or container, labelled **S**, provided at room temperature (see **Preparation of materials**)	at least 200 cm^3
Pieces of potato in a beaker or container, labelled **P** (see **Preparation of materials**)	at least 7 pieces
Distilled water in a beaker or container, labelled **W**, provided at room temperature	at least 200 cm^3
Methylene blue solution, labelled **M**, provided at room temperature (see **Preparation of materials**)	at least 15 cm^3
10 cm^3 or 20 cm^3 syringes, with the means to wash them out	2
2 cm^3 or 3 cm^3 syringe, with the means to wash it out	1
Pipettes, plastic or glass with a teat	2
Beakers or containers (capacity 75 cm^3 to 100 cm^3)	5
Test-tubes – large (to hold more than 25 cm^3 but no more than 50 cm^3)	5
Test-tube rack to hold 5 large test-tubes	1
Test-tubes – small (capacity 20 cm^3 to 30 cm^3)	5
Test-tube rack to hold 5 small test-tubes	1
Glass rod	1
Ruler, marked in mm	1
Scalpel or sharp blade	1
White tile or surface for cutting	1
Container with tap water (capacity approximately 200 cm^3), labelled **For washing**	1
Container (capacity approximately 200 cm^3), labelled **For waste**	1
Paper towels	8
Glass marker pen	1
Stop-clock or timer showing seconds	1

Question 2

1 DCPIP
2 Vitamin C solution
3 Lemon juice
4 Orange juice
5 Grapefruit juice
6 **6** Test tubes
7 **1** test tube rack
8 **3** 2-ml syringes
9 **2** 5-ml syringes
10 **1** Glass rod
11 **4** teat pipettes (droppers)
12 **4** 50-ml beakers
13 **1** stopwatch
14 **1** wash bottle labelled as **distill water**
15 **1** permanent marker
16 **1** white tile

Question 3

1 Microscope with an eyepiece graticule fitted into the eyepiece lens
For each candidate:
• the microscope must be set up on low power
• the slide must not be left on the stage of the microscope.

2 Slide **M1 –** Ranunculus *t.s.* root metaxylem